

Experiment 5

Problem:

Find factorial of a number using iterative and recursive functions.

Student Details:

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Code (C):

```
#include <stdio.h>
long long fact_iter(int n){
    long long res=1;
    for(int i=2;i<=n;i++) res *= i;
    return res;
}
long long fact_rec(int n){
    if(n<=1) return 1;
    return n * fact_rec(n-1);
}
int main(){
    char name[]="madhiya zaid javed"; int roll=36; char uin[]="251M040"; char divc='F';
    int n;
    printf("%s Roll:%d UIN:%s Div:%c\n",name,roll,uin,divc);
    printf("Enter a non-negative integer: ");
    if(scanf("%d",&n)!=1) return 0;
    printf("Iterative factorial of %d = %lld\n", n, fact_iter(n));
    printf("Recursive factorial of %d = %lld\n", n, fact_rec(n));
    return 0;
}
```

Sample Input and Output (example run):

Input: 5

Output: Iterative factorial of 5 = 120

Recursive factorial of 5 = 120